

When Should a Crop Map Say Less? Comparing Fixed Output Policies Under Unspecified Losses

Anonymous authors

Affiliation withheld for double-blind review

Abstract. Crop classifiers need not always emit one leaf label: a system can shrink its catalogue, abstain, return a label set, or back off to a coarser agronomic category. We compare fixed post-training policies from these four families within one seven-outcome account. Six cells carry loss; the representation is exact for losses that are constant within this predeclared basis, not for every possible downstream utility. Operating points are fixed on training and validation data, and the held-out block estimates the complete outcome-frequency vector of each policy. These vectors identify uniform dominance and linear switching boundaries without choosing a stakeholder’s loss. Across five frozen predictors on the reference-eligible PASTIS-R cohort, no within-predictor policy pair satisfied the dominance criterion over the whole declared loss simplex, while four policy types were preferred somewhere in the declared sensitivity sections. BreizhCrops supplies descriptive profiles but not paired inference under the frozen clustering unit. The study is exploratory and dataset-conditional: it does not claim geographic transport or one universally best response to uncertainty. Its contribution is a reproducible way to expose when the preferred output policy is determined by evidence and when it is determined by cost.

Keywords: Crop mapping · Selective classification · Set-valued prediction · Decision theory · Satellite image time series

1 Introduction

A parcel-level crop map is a promise: an inspector, adviser, or agency may act on every label. When a classifier is uncertain, answering anyway transfers that uncertainty downstream without showing it. Yet a map can promise less in several ways: shrink its class catalogue, abstain on a parcel, return a set of labels, or fall back to a coarser agronomic category. Each action trades a different kind of error against a different kind of incomplete answer.

The component ideas are established. Chow introduced the error–reject trade-off [5]; Ha generalized it to class-selective rejection [12]; and later work characterizes optimal set-valued decisions [17,6]. In crop mapping, catalogue size is already a measured design variable [9], and uncertainty-based rejection has been evaluated on the same benchmark [19]. What remains unresolved is how to compare these devices *with one another*: each normally fixes its own objective and reports its own curve.

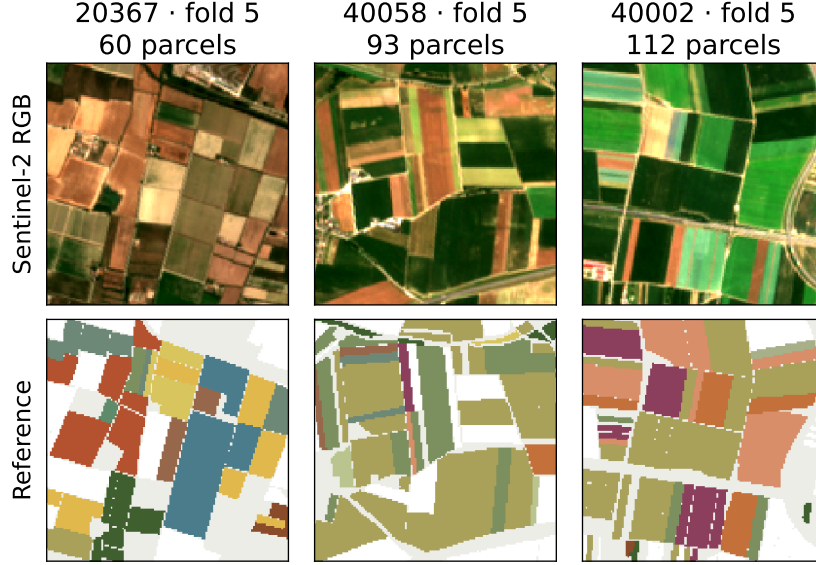


Fig. 1. Three real PASTIS-R patches in Sentinel-2 natural colour and their parcel-level reference labels. A frozen density-and-class-variety rule selected the panels before this revision; they provide visual context and are not model examples.

Within the bounded search in Section 2, we found no comparison that places fixed instances of catalogue shrinking, abstention, set-valued prediction, and taxonomic back-off in one decision problem and asks what follows without choosing the user’s loss. We address that narrower question with three contributions. First, we map every policy to a complete seven-cell outcome profile over one common parcel population. Second, we derive uniform dominance and switching regions over a predeclared six-cell normalized loss basis rather than selecting a favourable loss point. Third, we make the comparison auditable: operating points are fixed from training and validation, uncertainty is clustered at patch level, and outcome distributions and seed sensitivity remain visible.

The resulting claim is deliberately conditional. The data identify outcome frequencies and the loss regions in which a policy would be preferred; they do not identify stakeholder preferences. Both available parcel datasets had been inspected and remain exploratory, and the recorded search found no separate admissible confirmatory bank. We therefore make no transport claim and no universal recommendation: this paper is a reproducible decision atlas showing where the answer changes, not an assertion that one policy is best everywhere.

2 Related Work

We organize prior work by the limitation that leaves our comparison open.

Reject and set-valued decisions. Chow’s optimal error–reject rule treats silence as the alternative to one label [5]; Ha extends the choice to subsets of classes [12]. Mortier et al. optimize expected utility as a function of set size [17], while Chzhen et al. systematize these formulations and their trade-offs [6]. Ha’s class-selective rule is an implemented parameterized literature family, not part of the already executed six-arm family; a prospective exploratory extension fixes one validation-derived member without making the comparison confirmatory.

Learned abstention is developed, and out of our scope by choice. Deep selective classification supplies guarantees and learned rejectors [7,8], wagering losses [15], and AUC-based selection [18]. We do not reimplement them and we do not benchmark against them: we compare fixed abstention with three other output families. The open issue is between actions, not how to learn a better rejector.

Coverage, risk, and partial rejection. Adaptive conformal classification targets marginal and approximate conditional coverage [20]; RAPS regularizes unlikely labels to reduce set size at fixed coverage [1]. Risk-controlling prediction sets instead certify a bounded, user-specified loss at a chosen risk level [3]. Partial reject options optimize decisions from a firm label through set-valued and full rejection under a given reward [14]. Each fixes coverage or a loss specification within one output family. We leave the six loss coordinates unspecified and report where preference changes across families.

Earth observation and crop mapping. The closest precedent rejects uncertain pixels on PASTIS and makes the precision–recall trade-off explicit [19], but does not compare output families or their loss-space boundaries. Related rejection studies concern hazards [11], airborne vegetation mapping [10], or unknown-crop detection [4]. Catalogue size is already measured as a land-cover design variable [9]; thus catalogue shrinking is not our invention. Likewise, confidence-dependent crop hierarchies [23], hierarchical information fusion [2], and hierarchical prediction sets [16] precede our back-off policy. Our contribution is the shared outcome account used to compare these established actions.

Distribution and bounded novelty. Selective classification can magnify group disparities [13]; we use that result as motivation to retain classwise outcomes, not as evidence about crops. Within our search, we found no work comparing catalogue shrinking, per-parcel abstention, set-valued prediction, and taxonomic back-off as actions of one decision problem over a predeclared six-cell loss basis. The claim covers 61 automated and 6 manual English-language queries in arXiv, OpenAlex, and Semantic Scholar for 2019–2026, plus classics retrieved by name. It is not exhaustive: remote-sensing proceedings without DOIs are poorly indexed, and our selective-classification vocabulary missed the closest precedent, which we found by another route.

3 Problem Statement and Estimand

3.1 One Action Space

Let $\mathcal{Y}$ be the leaf catalogue, x_i the observations for parcel i , and $y_i \in \mathcal{Y}$ its reference label. Every evaluated policy returns

$$C_m(x_i) \subseteq \mathcal{Y}, \quad C_m(x_i) = \emptyset \text{ permitted.} \quad (1)$$

Catalogue shrinking. A subcatalogue is fixed once; the policy is silent when the unrestricted posterior maximizer falls outside it.

Per-parcel abstention. The maximizer is emitted only above a fixed confidence.

Set-valued prediction. A calibrated non-empty posterior prefix is emitted.

Taxonomic back-off. The policy emits the leaves below a named hierarchy node. The policies differ in which subsets are admissible and what fixes them, not in their output type. Two degenerate elements of the same space bound it: full-catalogue singleton prediction is the evaluated reference, while universal non-delivery is only a boundary condition.

3.2 Outcome Profile and Loss Space

Fix a dataset, region, campaign, split, predictor, and policy m at operating point τ_m . A predeclared map π assigns each parcel-level delivery to one of seven mutually exclusive outcomes: correct or incorrect precise delivery, non-delivery, a label set that contains or excludes the truth, or a coarse answer that contains or excludes it. Correct precise delivery is the zero-loss reference R_0 ; the remaining cells are $L_1, \dots, L_6$. The empirical object is the complete frequency profile

$$q_{m,j} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{O_i(m) = L_j\}, \quad j = 1, \dots, 6, \quad (2)$$

where N includes every eligible parcel, including non-deliveries. A delivered-only denominator would let a policy improve by answering less.

For any finite non-negative loss constant within these six cells, common positive scale does not affect ordering. We therefore use

$$\ell \in \Delta^5 = \{\ell \in \mathbb{R}_+^6 : \sum_j \ell_j = 1\}, \quad R_m(\ell) = \mathbf{q}_m^\top \ell. \quad (3)$$

This basis deliberately preserves the kind of promise but forgets distinctions within a cell, such as set cardinality, taxonomic depth, or which wrong crop was named. Expected set size is a descriptor, not the common loss: it charges equally for equally sized sets with different truth content and gives the empty set size zero. Uses that price the forgotten distinctions require an expanded basis.

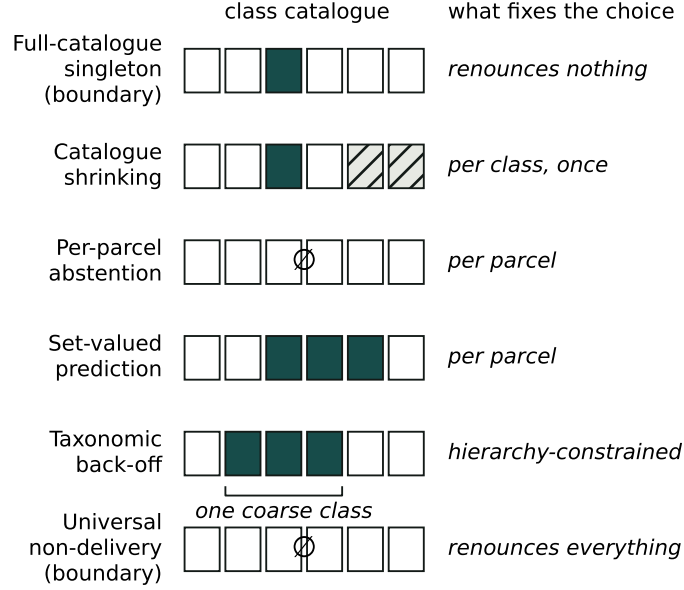


Fig. 2. All policies return subsets of one leaf catalogue, with the empty set allowed. The cells are structural and do not represent measured classes or frequencies.

The data determine $\mathbf{q}$, not ℓ . Policy m uniformly dominates n exactly when $q_{m,j} \leq q_{n,j}$ in every cost-bearing cell and at least one inequality is strict. Otherwise their switching boundary is $(\mathbf{q}_m - \mathbf{q}_n)^\top \ell = 0$. A plotted plane fixes the other coordinates and is a sensitivity section, not the full simplex or a context-independent one-number rule.

Operating points are fitted from training and validation only. Evaluation labels enter only after all deliveries exist; neither thresholds nor target rates are rematched on the held-out block.

3.3 Scope and Dependence

The estimand is conditional on each dataset, region, campaign, split, and predictor. The parcel is the observational unit and the image patch is the dependence cluster. Primary uncertainty resamples whole patches within the held-out spatial block; a separate sensitivity repartitions the same patches and reruns the complete family. Below three unique paired clusters, profiles stay descriptive and no interval or dominance claim is made.

Datasets are never pooled. The second bank is an external-region descriptive contrast, not another replicate of the first, and spatial subdivisions are sensitivity

groupings rather than independent territories. Consequently, we do not claim geographic transport: every result applies only to its named bank and design.

4 Fixed Policies as Alternative Actions

The main analysis instantiates fixed points from four post-training policy families as six arms. They do not cover the lattice. Cascades alter information and computation; class-specific thresholds add sparse validation-tuned parameters; learned rejectors alter training. Learned, class-adaptive, and multi-stage policies remain unexplored, so claims are restricted to these four points and the two catalogue rankings. Ha’s class-selective rule is an implemented parameterized literature family, not part of the already executed six-arm family. A prospective exploratory extension fixes one validation-derived member. Per model-stratum, its cost is the smallest second-highest validation posterior attaining the singleton target; it is diagnostic rather than an admission test.

4.1 Common Interface and Fixed Operators

Every arm returns a Boolean membership vector over the same leaf catalogue. One true entry is a precise delivery, several entries form an ambiguous or coarsened delivery, and an all-false row is non-delivery. The representation contains neither the reference label nor a delivered-only filter, so abstained parcels cannot disappear before evaluation.

The baseline emits the posterior maximizer. Each catalogue arm retains a different predeclared training-side class prefix and emits the unrestricted maximizer only when it belongs to that prefix. Abstention applies a validation-derived confidence threshold. Adaptive prediction sets sort labels by posterior and emit the smallest non-empty prefix reaching a calibrated mass. Calibration uses the exact split-conformal rank $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$; a rank beyond the calibration sample yields the full catalogue. The construction is deterministic, and held-out containment is reported rather than used to recalibrate it.

Taxonomic back-off keeps a confident singleton and otherwise emits the represented leaves below a predeclared HCAT4 node. The mapping resolves every leaf and cut node against EuroCrops and rejects missing, repeated, non-ancestral, or out-of-catalogue assignments. A cut node with one represented leaf remains a visible singleton; it is not widened after outcomes are seen.

All retained catalogues, thresholds, calibrated masses, and hierarchy cuts are inputs. The fitting interface accepts training and validation data but no held-out block; application accepts held-out posteriors but no labels; only the evaluator receives labels after every membership matrix exists. A fitted operating point stores row identities and rejects any training-validation overlap or reuse in evaluation, including copied rows. Thus test-label invariance belongs to the operator, while split disjointness is enforced at orchestration time.

The leaf catalogue follows the model output. A neutral evaluation universe is selected once from training support; validation and evaluation labels cannot

change it. Every arm retains every eligible parcel in the denominator. A selected class absent from evaluation remains visible with zero support and an undefined classwise statistic rather than being dropped or converted to a valid zero.

4.2 Closed Outcome Accounting

The evaluator maps each parcel to exactly one of the seven cells defined in Section 3.2, according to the delivery visible to the user. A singleton from the set arm is a precise answer; a multi-label answer remains distinct from a named coarse category; the empty row is non-delivery under every arm. The two failure cells for non-singleton answers prevent sets and coarse labels from being charged only when they contain the truth.

The primary interface returns the complete frequency vector and accepts no loss point. A separate sensitivity interface accepts six finite non-negative losses, fixes correct precise delivery at zero, and rejects incomplete mappings. Synthetic points verify that changing relative prices can reverse the preferred action; they are not empirical preferences. Expected set size remains a descriptor and does not rank the arms.

5 Evaluation Protocol

5.1 Banks, Splits, and Fixed Policies

We use optical PASTIS-R [22] and BreizhCrops [21]. The reference-eligible PASTIS-R fold-five cohort has 16640 parcels. Its four leading classes contain 72.0% of parcels and the six rarest contain 6.5%. These are parcel counts, not area shares; Fig. 3 is schematic.

Training fixes the class universe and catalogue rankings; validation fixes all other operating points. Stable parcel identities enforce disjointness before evaluation. The catalogue size is the smallest training-support prefix reaching 0.9 cumulative mass; abstention targets 0.9 delivery, the adaptive set uses 0.1 mis-coverage, taxonomic back-off targets 0.9 precise leaves, and the neutral universe requires 20 training parcels. These are author-chosen intervention points, not equal-loss points, and are never rematched on evaluation.

All arms are paired by parcel identity, block, and `patch_id`; reordering is allowed but omission, duplication, and substitution are not. Both banks had been inspected and remain exploratory. A closed register admitted no separate confirmatory bank, a bounded conclusion over its recorded candidates rather than an exhaustive public-data claim.

5.2 Uncertainty and Simultaneous Dominance

For each arm pair and outcome coordinate, the contrast is the left-arm indicator minus its comparator. The primary view resamples whole `patch_id` groups within spatial blocks, keeps all arms paired, and combines blocks with their original parcel shares. A sensitivity view instead gives equal weight to paired block

Four crops fill most of the mosaic
one cell = one parcel; schematic position

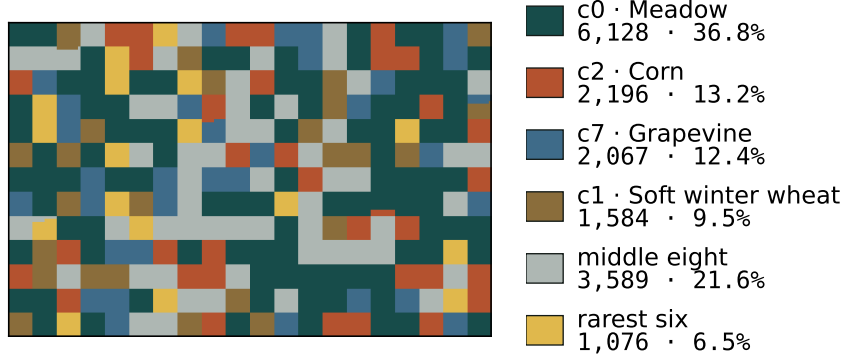


Fig. 3. Exact class support in the reference-eligible PASTIS-R fold-five cohort. Each micro-cell is one parcel; position and shape have no geographic or area meaning.

Table 1. Primary fixed policies. Each is one operating point, not a complete family.

Policy	Fixed before evaluation	Visible response
Singleton baseline	Posterior argmax	One leaf label
Support catalogue	Training-support prefix	One label or silence
F-score catalogue	Same size, frozen ordering	One label or silence
Abstention	Validation confidence threshold	One label or silence
Adaptive set	Split-conformal mass	Non-empty label set
HCAT4 back-off	Validation threshold and hierarchy	Leaf or named coarse class

means. Below three paired units, deltas remain descriptive and no interval, p -value, or multiplicity claim is returned.

For an ordered pair, $D_{m,n} = \max_j(q_{m,j} - q_{n,j})$. Inferential dominance requires every simultaneous one-sided bound to be at most zero and at least one to be strictly below zero. The frozen procedure uses a family-wise alpha of 0.05, 95% intervals, 9,999 paired patch-cluster resamples, add-one probabilities, and seed 20260909. The maximum statistic spans all 15 unordered arm pairs and 6 cost coordinates in a stratum. Missing comparisons are not removed after inspection; they withhold the affected claim.

A zero standard error normally withholds inference. The exception is structural: if neither action can produce an outcome, their difference is exactly zero in every cluster and its bound is the exact zero. At least one non-structural coordinate must improve strictly. Every resample recomputes its own block-centred

patch-cluster standard error; a cross-predictor claim must hold in the same direction for every predeclared predictor.

The winner region of arm m intersects the simplex with $(\mathbf{q}_m - \mathbf{q}_n)^\top \boldsymbol{\ell} \leq 0$ for every comparator. Two-dimensional sections declare all fixed coordinates and carry no probability measure. Point losses cannot choose operating points, define the primary family, or support stakeholder claims.

6 Results

We report exploratory outcome accounting, not a universal preference. Table 2 compresses the empirical object behind every comparison before any loss is applied. Its ranges are component-wise across the fixed predictor panel, not one synthetic profile. Figure 4 then exposes every joint profile: abstention moves mass from wrong precise labels to silence; adaptive sets move most mass to truth-containing ambiguous answers; and back-off introduces both correct and incorrect coarse deliveries. None of those moves has a sign until its cells receive a loss.

Table 2. Component-wise PASTIS-R outcome-frequency ranges across the fixed predictor panel (percent). R_0 is exact delivery; L_1 – L_6 are the declared cost-bearing cells. Endpoints vary independently, so a row is not one realizable joint profile; the structured artifact names every endpoint predictor.

Policy	R_0	L_1	L_2	L_3	L_4	L_5	L_6
Singleton	55.4–88.1	11.9–44.6	0.0	0.0	0.0	0.0	0.0
Support catalogue	55.1–81.4	10.2–43.8	1.1–8.4	0.0	0.0	0.0	0.0
F1 catalogue	55.3–77.4	7.8–43.9	0.8–15.1	0.0	0.0	0.0	0.0
Abstention	52.5–81.9	5.5–35.3	11.1–12.6	0.0	0.0	0.0	0.0
Adaptive set	2.7–6.8	0.1–0.3	0.0	90.9–92.2	1.3–6.1	0.0	0.0
Taxonomic back-off	53.8–84.1	7.3–38.5	0.0	0.0	0.0	3.4–5.4	3.2–4.3

For the 5 frozen PASTIS-R predictors, the primary atlas certified no dominance: 71 of 75 action pairs were certainly incomparable and 4 undetermined, not tied. This absence of dominance held in all ten seed runs. The historical finding that no spatial partition was selected failed in every seed run and is withdrawn; lacking per-seed selection profiles, we report no replacement.

No preregistered loss point exists to mark: point-loss selection is prohibited. Across the declared planar sections, the singleton baseline, adaptive sets, abstention, and taxonomic back-off each won somewhere; neither catalogue policy won a sampled point. This establishes switching on the declared grid, not a probability of being best or a volume of the simplex.

Figure 5 shows the first section of the frozen exhaustive enumeration, with $L_4 = L_5 = L_6 = 0.1$ and no pooling across predictors. It was selected for display after materialization, not preregistered. The 20 declared two-dimensional

DeepLabv3+							
singleton	64.9	35.1	0.0	0.0	0.0	0.0	0.0
support catalogue	63.4	31.7	4.8	0.0	0.0	0.0	0.0
F1 catalogue	63.5	30.7	5.8	0.0	0.0	0.0	0.0
abstention	62.2	26.7	11.1	0.0	0.0	0.0	0.0
label set	5.8	0.3	0.0	91.1	2.8	0.0	0.0
back-off	62.9	28.5	0.0	0.0	0.0	4.3	4.3
SegFormer							
singleton	64.7	35.3	0.0	0.0	0.0	0.0	0.0
support catalogue	63.5	33.0	3.5	0.0	0.0	0.0	0.0
F1 catalogue	63.2	32.0	4.7	0.0	0.0	0.0	0.0
abstention	61.7	27.0	11.3	0.0	0.0	0.0	0.0
label set	5.5	0.1	0.0	92.2	2.2	0.0	0.0
back-off	62.3	28.5	0.0	0.0	0.0	5.3	3.9
TSViT-Pheno							
singleton	85.8	14.2	0.0	0.0	0.0	0.0	0.0
support catalogue	79.9	12.4	7.7	0.0	0.0	0.0	0.0
F1 catalogue	77.4	10.5	12.1	0.0	0.0	0.0	0.0
abstention	80.6	7.6	11.8	0.0	0.0	0.0	0.0
label set	6.8	0.1	0.0	91.6	1.5	0.0	0.0
back-off	82.3	9.5	0.0	0.0	0.0	5.0	3.2
TSViT-FullIM							
singleton	88.1	11.9	0.0	0.0	0.0	0.0	0.0
support catalogue	81.4	10.2	8.4	0.0	0.0	0.0	0.0
F1 catalogue	77.1	7.8	15.1	0.0	0.0	0.0	0.0
abstention	81.9	5.5	12.6	0.0	0.0	0.0	0.0
label set	6.7	0.1	0.0	91.9	1.3	0.0	0.0
back-off	84.1	7.3	0.0	0.0	0.0	5.4	3.2
U-Net							
singleton	55.4	44.6	0.0	0.0	0.0	0.0	0.0
support catalogue	55.1	43.8	1.1	0.0	0.0	0.0	0.0
F1 catalogue	55.3	43.9	0.8	0.0	0.0	0.0	0.0
abstention	52.5	35.3	12.2	0.0	0.0	0.0	0.0
label set	2.7	0.3	0.0	90.9	6.1	0.0	0.0
back-off	53.8	38.5	0.0	0.0	0.0	3.4	4.3
	R0	L1	L2	L3	L4	L5	L6
	correct						

Fig. 4. Complete PASTIS-R outcome-frequency profiles for every frozen predictor and primary policy. Cell labels are percentages of the same reference-eligible cohort. Shade repeats the printed magnitude and is not the sole information channel. Columns are the zero-loss reference followed by the six cost-bearing cells defined in Section 3.2.

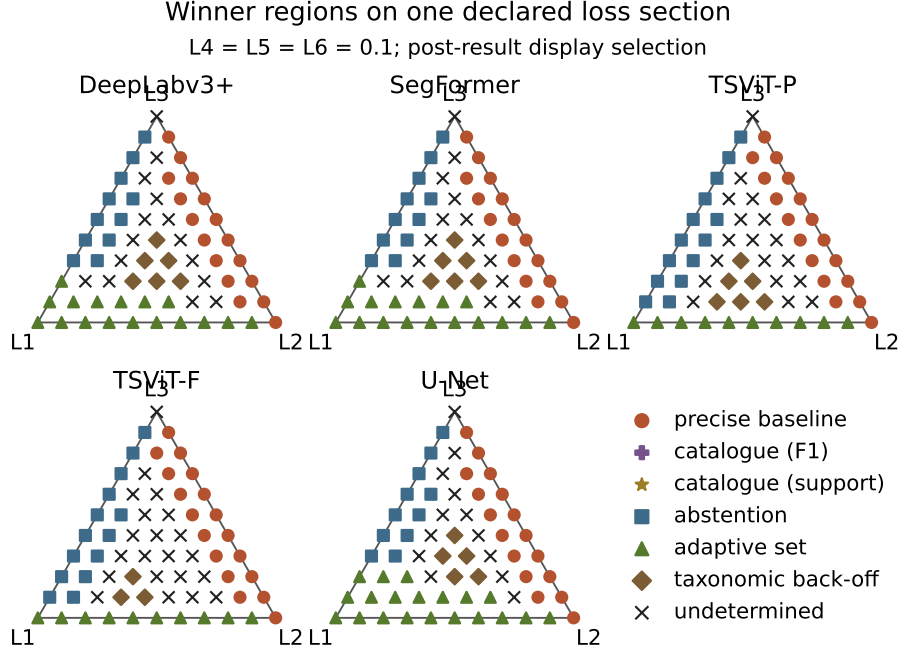


Fig. 5. Certain winner regions on one declared section of the exploratory PASTIS-R loss atlas. Symbols identify the winning action; crosses are evaluations for which the simultaneous band does not establish one winner. Each panel is a frozen predictor. Catalogue-restriction actions remain in the legend even though neither wins on this section, closing the six-action account. The displayed section is the first in the frozen enumeration and was selected for presentation after results.

sections contain 66 lattice points each. All 20 sections remain available in the structured artifact.

The conformal mass and the HCAT4 precise-delivery threshold were fitted on the frozen validation split. Both deliveries were fixed before evaluation labels were opened; reference availability only restricted the already-fixed PASTIS-R deliveries. Their validation targets are different operating constraints, not an equal loss.

Every conformal row met its marginal containment target, but only by returning multi-label answers. HCAT4 recovered additional truth-containing deliveries beyond correct precise answers in every row, yet also emitted wrong coarse answers in every row. Its realized precise-delivery rate varied across held-out strata despite one validation target. Thus a named coarse answer is neither a free rescue nor the same object as an arbitrary truth-containing set.

Marginal conformal calibration did not imply class-conditional coverage: one PASTIS-R U-Net class contained the truth in 28 of 206 sets. The historical

extreme BreizhCrops support was 6 parcels; seed-grid extrema spanned 1–6 and cannot sustain a class-level generalization.

Rows use reference-eligible evaluation cohorts: 16640 valid-label PASTIS-R parcels and 30000 BreizhCrops parcels per external-region stratum. The other 11892 PASTIS-R posterior identities lack a valid semantic reference for this outcome account. Delivery was fixed before that reference filter was applied, so the filter cannot improve a policy’s delivery rate; those identities are not silently described as evaluated parcels.

Together, the profiles and atlas answer the narrower question supported by the data: which policy is preferred for a declared six-cell loss, and where that preference changes. They do not identify the loss itself.

7 Discussion

The diagnostics expose why one success rate cannot answer the decision question. Conformal sets attain marginal containment by returning multiple labels; HCAT4 sometimes recovers the truth through a communicable parent and sometimes returns the wrong parent. Calling either uniformly better would silently price breadth or coarse error at zero, so the outcome coordinates must remain separate.

The coverage statement is conditional on each evaluated bank and on exchangeability between its calibration and evaluation parcels. Both banks had been inspected before this execution, so every result here is exploratory even though the evaluation partitions are held out from calibration. No transport claim follows, and the closed dataset search did not find a separate admissible confirmatory benchmark. Predictor-to-predictor variation is sensitivity, not a route to select a favourable model. PASTIS-R therefore supports an exploratory decision atlas; BreizhCrops supports descriptive outcome profiles but not paired inference because its frozen reporting structure has only 1 cluster per model–region stratum.

No certified dominance is not a tie: crossing profiles and wins for four actions reject a price-free ranking while locating its switches. Catalogue restriction winning no sampled point does not prove an empty winner region because the grid has no simplex measure. The spatial grouping conclusion changed under

Table 3. Inferential scope by bank. Roles are frozen and the banks are never pooled.

Bank	Permitted role	Evidence used here
PASTIS-R	Exploratory	Dataset-conditional profiles and patch-cluster atlas
BreizhCrops	Descriptive only	External-region profiles; one cluster per model–region stratum, so no paired intervals

the frozen seed grid, so we withdraw it rather than choose a favourable partition after seeing the atlas.

Two scope restrictions are load-bearing. First, each arm is one fixed operating policy; the study does not order every possible member of its family. Second, the six-cell account intentionally forgets set cardinality, taxonomic depth, and which crop pair was confused. A downstream use that prices any of those distinctions requires an expanded outcome basis before applying the same geometry.

The inherited PASTIS-R checkpoints do not record training seeds in their declared identity sources. Seed sensitivity can therefore rerun only post-training stochastic stages for those members; it cannot establish robustness to training randomness. We do not backfill that claim by retraining after seeing the atlas. Stable predicates are retained; changed counts above are ranges. BreizhCrops sampling and classical fits remain rerunnable and separate, without pooling.

Label-noise robustness is likewise unmeasured: no transition model tailored to either bank was declared, and the withheld secondary family supplies no estimable corruption target. We therefore do not inject a convenient synthetic noise process after seeing the results.

Data and Code Availability

The anonymous source archive contains the L^AT_EX project, bibliography, generated appendix, and figures, but no raw satellite data. A separate anonymous supplement reconstructs all seventy-five pairwise atlas verdicts and the declared lattice from aggregate patch counts; it contains no imagery, parcel records, repository access, or author metadata. Training artifacts remain private during review and will accompany a citable archive before publication. Source data retain their providers' terms; readers re-run the atlas and figures from outcome counts, not imagery.

References

1. Angelopoulos, A., Bates, S., Malik, J., Jordan, M.I.: Uncertainty sets for image classifiers using conformal prediction (2020)
2. Barriere, V., Claverie, M., Schneider, M., Lemoine, G., d'Andrimont, R.: Boosting crop classification by hierarchically fusing satellite, rotational, and contextual data. *Remote Sensing of Environment* **305**, 114110 (2024). <https://doi.org/10.1016/j.rse.2024.114110>
3. Bates, S., Angelopoulos, A., Lei, L., Malik, J., Jordan, M.: Distribution-free, risk-controlling prediction sets. *Journal of the ACM* (2021). <https://doi.org/10.1145/3478535>
4. Carvalho, T., Martinez, J.A.C., Oliveira, H., dos Santos, J.A., Feitosa, R.Q.: Outlier exposure for open set crop recognition from multitemporal image sequences. *IEEE Geoscience and Remote Sensing Letters* **20**, 1–5 (2023). <https://doi.org/10.1109/LGRS.2023.3244532>
5. Chow, C.: On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory* (1970). <https://doi.org/10.1109/tit.1970.1054406>

6. Chzheng, E., Denis, C., Hebiri, M., Lorieul, T.: Set-valued classification — overview via a unified framework (2021)
7. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: *Advances in Neural Information Processing Systems* 30 (NIPS 2017). pp. 4878–4887. Curran Associates (2017), <https://proceedings.neurips.cc/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html>
8. Geifman, Y., El-Yaniv, R.: SelectiveNet: A deep neural network with an integrated reject option. In: *Proceedings of the 36th International Conference on Machine Learning (ICML)*. *Proceedings of Machine Learning Research*, vol. 97, pp. 2151–2159. PMLR (2019), <https://proceedings.mlr.press/v97/geifman19a.html>
9. Ghassemi, B., Izquierdo-Verdiguier, E., d’Andrimont, R., Vuolo, F.: How accurately and in what detail can land use and land cover be mapped using copernicus Sentinel and LUCAS 2022 data? *Remote Sensing* **17**(8), 1379 (2025). <https://doi.org/10.3390/rs17081379>
10. Gimenez, R., Laloue, A., Fabre, S.: Rejection methods for vegetation mapping using hyperspectral airborne data. *International Journal of Remote Sensing* **44**(16), 4937–4962 (2023). <https://doi.org/10.1080/01431161.2023.2240520>
11. Gonzalez-Calabuig, M., Cohrs, K.H., Nedungadi, V., Osika, Z., Cartuyvels, R., Knoblauch, S., Massant, J., Nath, S., Ebel, P., Sitokoustantinou, V.: SHRUG-FM: Reliability-aware foundation models for Earth observation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. pp. 7919–7928 (2026)
12. Ha, T.M.: The optimum class-selective rejection rule. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **19**(6), 608–615 (1997). <https://doi.org/10.1109/34.601248>
13. Jones, E., Sagawa, S., Koh, P.W., Kumar, A., Liang, P.: Selective classification can magnify disparities across groups. In: *9th International Conference on Learning Representations (ICLR)* (2021), https://openreview.net/forum?id=NOM_4BkQ05i
14. Karlsson, M., Hössjer, O.: Classification under partial reject options. *Journal of Classification* (2023). <https://doi.org/10.1007/s00357-023-09455-x>
15. Liu Ziyin, Wang, Z., Liang, P.P., Salakhutdinov, R., Morency, L.P., Ueda, M.: Deep gamblers: Learning to abstain with portfolio theory. In: *Advances in Neural Information Processing Systems* 32 (NeurIPS 2019). pp. 10622–10632. Curran Associates (2019), <https://proceedings.neurips.cc/paper/2019/hash/0c4b1eeb45c90b52bfb9d07943d855ab-Abstract.html>
16. Mortier, T., Hüllermeier, E., Dembczyński, K., Waegeman, W.: Set-valued prediction in hierarchical classification with constrained representation complexity. In: *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*. *Proceedings of Machine Learning Research*, vol. 180, pp. 1392–1401. PMLR (2022), <https://proceedings.mlr.press/v180/mortier22a.html>
17. Mortier, T., Wydmuch, M., Dembczyński, K., Hüllermeier, E., Waegeman, W.: Efficient set-valued prediction in multi-class classification. *Data Mining and Knowledge Discovery* **35**, 1435–1469 (2021). <https://doi.org/10.1007/s10618-021-00751-x>
18. Pugnana, A., Ruggieri, S.: AUC-based selective classification. In: *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*. *Proceedings of Machine Learning Research*, vol. 206, pp. 2494–2514. PMLR (2023), <https://proceedings.mlr.press/v206/pugnana23a.html>
19. Rey, M., Mnih, A., Neumann, M., Overlan, M., Purves, D.: Uncertainty evaluation of segmentation models for Earth observation (2025)

20. Romano, Y., Sesia, M., Candès, E.J.: Classification with valid and adaptive coverage (2020)
21. Rußwurm, M., Pelletier, C., Zollner, M., Lefèvre, S., Körner, M.: BreizhCrops: A time series dataset for crop type mapping. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* **XLIII-B2-2020**, 1545–1551 (2020). <https://doi.org/10.5194/isprs-archives-XLIII-B2-2020-1545-2020>
22. Sainte Fare Garnot, V., Landrieu, L., Chehata, N.: Multi-modal temporal attention models for crop mapping from satellite time series. *ISPRS Journal of Photogrammetry and Remote Sensing* **187**, 294–305 (2022). <https://doi.org/10.1016/j.isprsjprs.2022.03.012>
23. Turkoglu, M.O., D’Aronco, S., Perich, G., Liebisch, F., Streit, C., Schindler, K., Wegner, J.D.: Crop mapping from image time series: Deep learning with multi-scale label hierarchies. *Remote Sensing of Environment* **264**, 112603 (2021). <https://doi.org/10.1016/j.rse.2021.112603>